IC arity/oracle threshold

Independent review synthesis

2026-11-01 · BATCH-a68666 · TASK-20261101-bc06d9

Fresh machine verification passes. The named MITM inequality holds in its real cost model. The bundled claim needs revision for attainment and pointwise/optimized quantifiers.

1. Machine status and semantic fidelity

Hosted workflow **41275109694**, attempt 1, pins source `7e1827b530349c633d98c96185e8196669812cb4`. The native target receipt reports successful build and axiom audit, no forbidden constructs and no infrastructure failure. Its dedicated audit is:

For real $m \geq 2$, $\beta > 0$ and $k \geq m/2$:
$1/2 < \mathrm{costAll}(m, \beta, k)$.

The audit reports only propext, Classical.choice and Quot.sound. The 894-theorem project audit and built module do not certify the entire bundled English claim. Only the MITM declaration has the dedicated named-target audit.

Component	Machine status	Fidelity / outcome
Named MITM theorem	Passed	Faithful scoped; real-model inequality holds
Blanket attainment	Not the named declaration	Mismatch; explicit counterexample
Pointwise below-half iff	Rational thresholds only	Mismatch; requires optimization quantifier
Blind full envelopes	Not newly formalized	Symbolic derivation holds; complete domains
Algorithm / PFSE bridge	Not supplied	Open interfaces, not an algorithm theorem

Owned joints are not votes

A / TASK-20261101-bd53ca: breaks the exact bundled claim, while confirming the named theorem at model scope. **B / TASK-20261101-e90be2:** holds for a derivation from definitions alone; B did not inspect or approve the producer theorem. Their findings agree on nonattainment and on the need to distinguish quantifiers.

Internal sources: A review §§ Clause-to-declaration map, Explicit breaking artifacts; B derivation §§1–3,7. Full paths and SHA-256 bindings are in independence-check.json and report.md. No external literature claim is added.

2. Complete real-parameter envelopes

Domain: real $m \geq 2$, $\beta > 0$, $k \geq 0$. Write F = costAll, G = costOne.

$F = \max(\max(0, 1 - \beta(m-1)) + k\beta, 2\beta)$
$G = \max(\max(0, 1 - \beta m) + (1+k)\beta, 2\beta)$
$d = m-1-k$; $q_F = \max(k,2)$; $q_G = \max(k+1,2)$
$F = \max(1-d\beta, q_F\beta)$; $G = \max(1-d\beta, q_G\beta)$.

For either q, d+q > 0. The sign split is exhaustive:

Regime	Infimum	All minimizers β > 0
$k < m-1$	$q/(d+q)$	Unique $\beta = 1/(d+q)$
$k = m-1$	1	$0 < \beta \leq 1/q$
$k > m-1$	1	None; approach $\beta \to 0+$

Below $k=m-1$, F has value $2/(m+1-k)$ for $k\leq 2$ and $k/(m-1)$ for $k\geq 2$. G has value $2/(m+1-k)$ for $k\leq 1$ and $(1+k)/m$ for $k\geq 1$. Shared boundary formulas agree. Beyond $k=m-1$ the infimum is 1, **not** those rational branches.

Optimized versus pointwise

For either expression, its optimized value is below, equal to, or above 1/2 according as d is greater than, equal to, or less than q. For F, optimized below-half means $m>k+3$ when $k\leq 2$, or $2k<m-1$ when $k\geq 2$.

For a fixed β, when d>q, the value is below 1/2 exactly on $1/(2d)<\beta<1/(2q)$. When d=q it equals 1/2 only at $\beta=1/(2q)$. When d<q it exceeds 1/2 for every $\beta>0$.

$F\leq G$ pointwise; equality holds exactly for $k\leq 1$ or $\beta\leq 1/m$. Their optimized values are equal exactly for $k\leq 1$ or $k\geq m-1$. Strict optimized inequality requires $1<k<m-1$.

MITM scope

$k\geq m/2$ implies $d<q_F$, hence F>1/2 pointwise. B derives the sharp optimized lower bounds:

$I_F \geq 4/(m+2)$ for $2\leq m\leq 4$; $I_F \geq m/(2(m-1))$ for $m\geq 4$.
$I_G \geq 1/2+1/m$.

The gap above 1/2 can shrink as m grows; no positive uniform margin is asserted for unbounded m. A lower bound of 1/2 nevertheless persists in this model class. Source: B derivation §§1–5,8 (symbolic, not new Lean evidence).

3. Exact defects and nearby controls

These substitutions are copied/derived from the sealed reports, not numerical trials or newly compiled counterexamples.

Test	Result and scope
F1: m=2,k=2	F=max(1+β,2β), G=max(1+β,3β). Both exceed 1 for β>0 and tend to 1 as β→0+. Infimum 1; no minimizer. A and blind B agree.
F2: middle bound	m=5,k=0 gives lower bound 0, never attained. Actual minimum 1/3 occurs at β=1/6. Attainment needs the missing regime restriction.
F3: pointwise iff	m=5,k=1 satisfies m>k+3, but β=1 gives F=2; β=1/5 gives F=2/5. An optimized condition is not a pointwise iff. m=7,k=2,β=1 also gives 2.
F4: strictness	m=3,k=2 gives both optima 1. Difference (1+k)/m−k/(m−1)=(m−1−k)/(m(m−1)); strictness requires k<m−1.
F4: equality	m=5,k=2,β=1/4 gives F=1/2. “Never below” does not imply “cannot reach.”
F5: lost domain	m=2,k=3/2 gives unattained infimum 1, not the unqualified small-branch value 4/3. The theorem’s k≤m−2 condition fails.

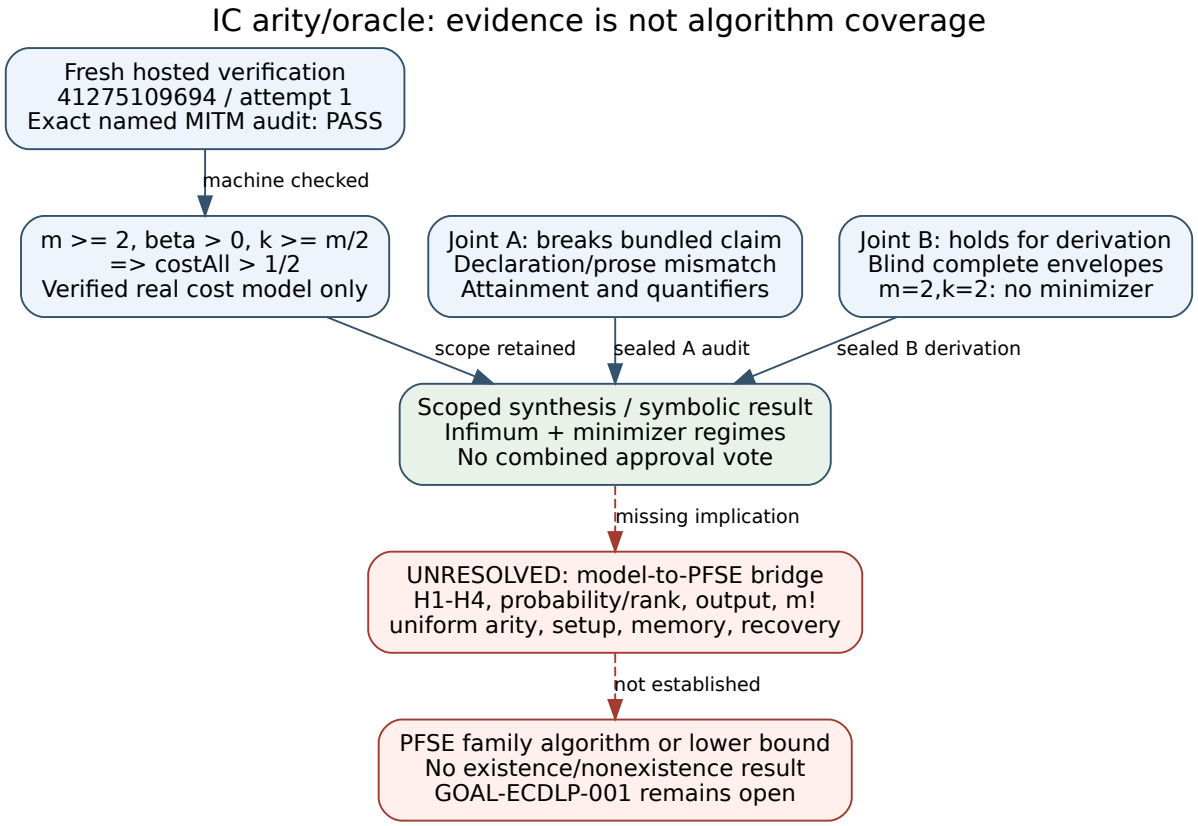
Planned controls retain changed hypotheses

- **Positive:** F(5,1/5,1)=2/5; an admitted subthreshold model oracle exists as a formula.
- **Boundary:** F(3,1/4,0)=1/2, not a strict tie at arbitrary β.
- **Drop H3:** attempts-only exponent 1/3 at m=3,k=0,β=1/3; full exponent 2/3.
- **Drop H1/H2:** modified costs can be 1/4 or 2/5; they are not original-model results.
- **Change H4:** at m=4, free oracle minimum 2/5 versus order-one minimum 1/2.
- **Exception scope:** there is no algorithm-to-H1–H4 coverage premise for the anomalous/pairing controls. No concrete curve fixture was run.

A’s full declaration/hypothesis-delta table records the missing k≥2 in the middle attainment clause, implied denominator positivity, fixed-parameter specializations and the extra optimization quantifier. The named MITM signature itself has no unexplained binder. H1–H4 are external modeling obligations, not hidden theorem assumptions.

Affected target/header/note/lead bytes remain frozen. The separate corrective task proposed on page 5 must record formalization errata and narrowed citations.

4. Formula-to-algorithm bridge remains explicit



Solid edges: cited machine check or sealed review. Dashed red edges: missing implications, not verified algorithm coverage. Editable source: dependencies.dot. No performance measurements are plotted.

Interface	Still required
H1 / H2	Yield distribution, hit probability, useful independent rank, target relations, stopping time and restarts
H3 / H4	Actual bit-cost LA; oracle return semantics, success, output enumeration/verification, achieved cost
Growing arity	m!, exact multiset correction, integer rounding, degree/row length and every m-dependent overhead uniformly
Full pipeline	Family measure/sampler; setup, preprocessing, transfer, descent, recovery, final verification, peak/persistent memory, processors and communication

For Y=L with probability μ/L, E[Y]=μ but Pr[Y>0]=μ/L: a mean does not give a success law. Returned relation rows may all be duplicates. Explicit output and compressed-output recovery must be priced. For growing integer m, dropping m! needs m log m=o(n). Fixed-m $r^{o(1)}$ terms are not automatically uniform in m=m(n).

With k=1, β=1/m, m>2 the formula gives F=G=2/m→0 as m grows. This does not implement a solver or supply probability, rank, descent or memory. The curve-algebra documentation constraint is retained: identities, conditional model costs and total algorithmic costs are different objects.

5. Custody, independence and one next action

The source review outputs are sealed at `76320e68aef505292da6a9eff98aa219cca4ecc3`. The full five-artifact snapshot is `5658c82b1de6f4fc6b44c864c91855c61c21d72e`, parent `446bf35095bf2744a817b0fad7dd3bd170bcb507`. Native archive checks pass every declared hash, ancestry, parent and all three IDs. The archived provisional pending-gate bytes remain unchanged; later finalization and lifecycle commit `14886d001e60d6bd6fdecd145e79c0153b56ee98` make the synthesis dependency complete.

The externally pinned plan SHA-256 is `2ce285e77cd85699f47a9ee4826775fdfc2a077168000fbae30e805b81813e92`. The checker composes one prospective addendum and returns:

> PASS: 2 reports; every joint owned and attested; blindness respected; controls declared. Full command/stdout/stderr and exact hashes: independence-check.json.

A and B attest no sibling access. B read only the three isolated allowed inputs; Coordinator compares now, after archive. Neither reviewer is unblinded. Actual serving-model/effort/fork settings and backend isolation remain unverified. The checker checks declarations, not their truthfulness.

The 67,574-byte target receipt is losslessly committed; native SHA-256: `75e52bc6bb7a034f81103166cf463d3f660b39b863177b797f7039f58394bc0a`. Fresh maps match 75 workspace and 10 verifier entries. Historical PR2854 receipt mismatch is not relabeled. GitHub artifact 13072015323 has finite expiry **2027-01-30T04:58:42Z**. Raw ZIP S3 archive is **unavailable_not_attempted**; temporary copies are nondurable. No permanent full-ZIP custody or independent kernel recheck is claimed.

Exactly one bounded proposed action

Versioned correction/formalization of the complete real-model target. Preserve frozen artifacts; record m=2,k=2 nonattainment in a dated correction, map affected target/header/note/lead citations, and specify/prove the d/q infimum, all minimizers and optimized/pointwise threshold regimes with F1–F5 and positive controls. Require later fresh verification and affected-joint fidelity review before any model-to-PFSE use. This packet launches none of it. See next-action.json.

Scope/provenance: internal packet only; no new primary-literature claim or experiment. No canonical DOT/SVG/Mermaid graph was present in the materialized study/frontier scope; PFSE model/cost tables remain unchanged. Initial Typst installation hit sandbox networking; an authorized network retry installed Typst 0.15.0 in scratch. Source and PDF are documentation, not a speedup or lower bound. Full report and exact controls: report.md. Visual check: visual-inspection.json. Final archival is TASK-20261101-7e3b01, not this task.

> No existence/nonexistence result is established. GOAL-ECDLP-001 remains open. No source/proof/status, EV/DEC/H, commit/push/release or final publication change is made by this synthesis.